UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF M.Sci. WITH HONOURS

FOURTH-YEAR FINAL EXAMINATION

03 01087

LM NUCLEAR PHYSICS

SUMMER EXAMINATION 2024

Time Allowed: 2 hours

Answer four questions from Section 1 and two questions from Section 2.

Section 1 consists of four questions and carries 40% of the marks for the examination. Answer ***all four*** questions from this Section.

Section 2 consists of three questions and carries 60% of the marks. Answer ***two*** questions from this Section. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

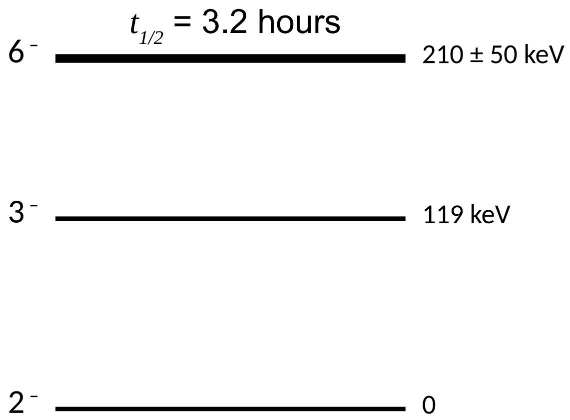
ANY CALCULATOR

SECTION 1

Answer **all four** *questions from this Section.*

1. What is the spin and parity, I^π, of the deuteron? Show the orbital angular momentum between the proton and neutron can take values $L = 0$ or 2.
   What are the consequences for the electric quadrupole moment of the deuteron? **[10]**

2. The radioactive isotope, ${}^{190}_{75}$Re, can exist in a long-lived excited state. The low-lying energy levels are shown below.



   The measured excitation energy of the metastable state is uncertain by $\pm$50 keV, such that it lies in the range 160 to 260 keV. Which transition from the metastable state has the lowest multipolarity?
   For this transition, estimate the range of conversion coefficients, for a K electron, due to the uncertainty in level energy.
   How could measuring the relative X-ray intensity be used to constrain the energy of the metastable state? **[10]**

3. (a) Demonstrate that, for the nucleon-nucleon interaction, both the tensor and spin-dependence components can arise from terms of the form, $\left(\underline{s_1} \times \underline{r}\right) \cdot \left(\underline{s_2} \times \underline{r}\right)$, where s_1 and s_2 are the spins of the two nucleons. Label the tensor term. [Hint: use the vector identity: $(\underline{A} \times \underline{B})(\underline{C} \times \underline{D}) = (\underline{A} \cdot \underline{C})(\underline{B} \cdot \underline{D}) - (\underline{B} \cdot \underline{C})(\underline{A} \cdot \underline{D})$.] **[4]**

   (b) What is meant by the term 'exchange force'?
   Draw a Feynman diagram showing the π^+ exchange between a proton and neutron. Show the quark content of each particle. **[6]**

4. What are the selection rules for allowed Fermi (F) and Gamow-Teller (GT) β-decays?
   Examine the β-decays for the nuclei shown below, and state whether they correspond to F, GT or if both decays are possible. State whether the decays are allowed, or by which order they are forbidden.
   Which nucleus has the shortest half life?

Initial state		Final state	
$^{61}_{27}$Co	$I^\pi = \frac{7}{2}^-$	$^{61}_{28}$Ni	$I^\pi = \frac{3}{2}^-$
$^{12}_{5}$B	$I^\pi = 1^+$	$^{12}_{6}$C	$I^\pi = 0^+$

[10]

SECTION 2

Answer **two** *questions from this Section. If you answer more than two questions, credit will only be given for the best two answers.*

5. (a) Explain all the processes that occur during electron capture by a nucleus. If both electron capture and β^+-decay are allowed, explain when electron capture dominates. **[4]**

(b) The isotope ^{211}At is unstable. Write the decay equations, and, using the data below, demonstrate which decay modes are energetically allowed by calculating the associated Q-values.
[$\Delta m(^{211}_{86}\text{Rn}) = -8.755$ MeV, $\Delta m(^{211}_{85}\text{At}) = -11.647$ MeV,
$\Delta m(^{211}_{84}\text{Po}) = -12.432$ MeV, $\Delta m(^{207}_{83}\text{Bi}) = -20.054$ MeV and
$\Delta m(^{4}_{2}\text{He}) = 2.424$ MeV.] **[8]**

(c) For electron capture Fermi's Golden Rule predicts that

$$\lambda = \frac{2\pi}{\hbar} V_{fi}^2 \frac{dn}{dE_f},$$

where, for electron capture,

$$\begin{aligned} V_{fi} &= \int \psi_f^* O_p \psi_i dV \\ &= \int \psi_{z-1}^*(r_{Z-1}) \quad \psi_\nu^*(r_\nu) O_p \psi_Z(r_Z) \psi_e(r_e) d^3r_{Z-1} d^3r_Z d^3r_e d^3r_\nu . \end{aligned}$$

Describe the four assumptions that lead to

$$V_{fi} = \frac{N^{3/2}}{\sqrt{\pi V}} g_F \int \psi_{Z-1}^*(r) \psi_Z(r) d^3r = \frac{g_F}{\sqrt{V\pi}} \left[\frac{Zme^2}{4\pi\varepsilon_0\hbar^2}\right]^{3/2} M_{fi},$$

and define the terms M_{fi} and g_F. **[10]**

(d) The decay constant for electron capture is

$$\lambda = \frac{Z^3 m^3 e^6}{32\pi^5 \varepsilon_0^3 \hbar^{10} c^3} |M_{fi}|^2 g_F^2 E_\nu^2 \, .$$

Calculate the half-life for the *super-allowed* electron capture of $^{14}_{8}$O. The measured half-life is 70.6 seconds. From the difference between the calculated electron capture half-life and the measured decay half-life, what do you conclude about the dominant decay mechanism?
[$\Delta m(^{14}\text{O}) = 8007.35596$ keV, $\Delta m(^{14}\text{N}) = 2863.41699$ keV, and
$g_F = 88$ eV fm^3] **[8]**

6. (a) Describe how the short-range repulsion of the nucleon-nucleon force can be understood from the properties of the quark-quark interaction. **[3]**

   (b) Describe three properties of the nucleon-nucleon interaction, excluding the short-range repulsion, that can be observed from nucleon-nucleon scattering. **[6]**

   (c) Alpha decay is mediated by the strong force. From the α-particle transmission probability,

$$T_\alpha = \exp\left\{-2\sqrt{\frac{2m}{\hbar^2 Q}}\frac{Z_1 Z_2 e^2}{4\pi\varepsilon_0}\left[\frac{\pi}{2} - 2\left(\frac{Q}{B}\right)^{1/2}\right]\right\},$$

   where symbols have their usual meaning. Show that the Geiger-Nuttall law,

$$\log_{10} t_{1/2} = C + \frac{D}{\sqrt{Q}}$$

   can be obtained, where C and D are constants. **[5]**

   (d) The frequency with which the α-particle hits the barrier, f, also affects α-particle emission. Estimate f stating any assumptions you make. **[4]**

   (e) i. Calculate the half-life for the α-decay of ${}^{174}_{72}$Hf to the ground-state of ${}^{170}_{70}$Yb.
   [$V_0 = 35$ MeV $\Delta m({}^{174}_{72}\text{Hf}) = -55.844$ MeV,
   $\Delta m({}^{170}_{70}\text{Yb}) = -60.763$ MeV and $\Delta m({}^{4}_{2}\text{He}) = 2.424$ MeV.] **[8]**

   ii. The first excited state in ${}^{170}_{70}$Yb lies at 0.084 MeV with $I^\pi = 2^+$. Would the half-life for the competing α-decay branch to the 2^+ state be longer or shorter than the half-life of the decay to the ground-state? Explain your answer. **[4]**

7. (a) i. Give two pieces of experimental evidence for shell structure in nuclei? **[2]**

ii. What is implied by the term 'mean field'? **[2]**

(b) i. The Woods-Saxon potential has the form,

$$V(r) = \frac{V_0}{1 + \exp\left[\frac{r-R}{a}\right]}.$$

State the physical meaning of each of the parameters and explain why a potential of this shape is used in the nuclear shell model. **[4]**

ii. Sketch and label a diagram of a Woods-Saxon potential. **[2]**

iii. Calculate the fractional change in the potential depth between the limits $r = R \pm a$. Give your answer in terms of V_0. **[4]**

(c) The sequence of energy levels in the shell model is

$$1s_{1/2}, 1p_{3/2}, 1p_{1/2}, 1d_{5/2}, 2s_{1/2}, 1d_{3/2}, 1f_{7/2}, 2p_{3/2}, 1f_{5/2} \ldots$$

Explain why there are two states for each orbital angular momentum, l, and obtain an expression for how the splitting between them depends on the magnitude of l. **[8]**

(d) In the shell model, the expectation value of the nuclear magnetic dipole moment is

$$\begin{aligned}\langle \mu \rangle = \frac{1}{2(j+1)} &\{g_l\left[j(j+1) + l(l+1) - s(s+1)\right] \\ + \ &g_s\left[j(j+1) + s(s+1) - l(l+1)\right]\}\,\mu_N,\end{aligned}$$

where the intrinsic-spin g-factors for free neutrons and protons are, $g_s^n = -3.826084$ and $g_s^p = 5.585691$. Use the above equation to calculate the magnetic dipole moments for:

i. ${}^{13}_{7}$N and,

ii. ${}^{17}_{9}$F. **[6]**

The measured values are $+0.3222\mu_N$ (^{13}N) and $4.7213\mu_N$ (^{17}F). What can you conclude from comparison with your calculations? **[2]**

Formula Sheet

Coulomb barrier, $V_{Coul} = \dfrac{Z_1 Z_2 e^2}{4\pi\varepsilon_0 R}$, where $R = 1.2A^{1/3}$ fm

Radial Schrödinger equation, $-\dfrac{\hbar^2}{2m}\dfrac{d^2u}{dr^2} + \left[V(r) + \dfrac{l(l+1)\hbar^2}{2mr^2}\right]u(r) = Eu(r)$,

where $R(r) = \dfrac{u(r)}{r}$

Deuteron wave function matching condition, $k_1 \cot(k_1 R) = -k_2$

where $k_1 = \sqrt{\dfrac{2m(E+V_0)}{\hbar^2}}$ and $k_2 = \sqrt{-\dfrac{2mE}{\hbar^2}}$

Magnetic moment, $\langle\mu\rangle =$

$$\frac{1}{2(j+1)}\left\{g_l\left[j(j+1)+l(l+1)-s(s+1)\right]+g_s\left[j(j+1)+s(s+1)-l(l+1)\right]\right\}\mu_N$$

Alpha-decay half-life, $t_{1/2} = \ln(2)\dfrac{R}{c}\sqrt{\dfrac{mc^2}{2(V_0+Q)}}\exp\{2G\}$,

where G is the Gamow factor, $G = \sqrt{\dfrac{2m}{\hbar^2 Q}}\dfrac{Z_1 Z_2 e^2}{4\pi\varepsilon_0}\left[\dfrac{\pi}{2} - 2\left(\dfrac{Q}{B}\right)^{1/2}\right]$

Beta-decay ft values, $f(Z, T_e)t_{1/2} = \ln(2)\dfrac{2\pi^3\hbar^7}{g_F^2|M_{fi}|^2 m_e^5 c^4}$

Internal conversion coefficients, $\alpha(ML) \approx \dfrac{Z^3}{n^3}\left(\dfrac{e^2}{4\pi\varepsilon_0\hbar c}\right)^4\left(\dfrac{2m_e c^2}{E}\right)^{L+3/2}$ and

$$\alpha(EL) \approx \frac{Z^3}{n^3}\left(\frac{L}{L+1}\right)\left(\frac{e^2}{4\pi\varepsilon_0\hbar c}\right)^4\left(\frac{2m_e c^2}{E}\right)^{L+5/2}$$

Weisskopf estimates (for E_γ in MeV),

$\lambda(E1) = 1.0\times10^{14}A^{2/3}E_\gamma^3$

$\lambda(M1) = 3.1\times10^{13}E_\gamma^3$

$\lambda(E2) = 7.4\times10^{7}A^{4/3}E_\gamma^5$

$\lambda(M2) = 2.2\times10^{7}A^{2/3}E_\gamma^5$

$\lambda(E3) = 3.5\times10^{1}A^{2}E_\gamma^7$

$\lambda(M3) = 1.1\times10^{1}A^{4/3}E_\gamma^7$

$\lambda(E4) = 1.1\times10^{-5}A^{8/3}E_\gamma^9$

$\lambda(M4) = 3.3\times10^{-6}A^{2}E_\gamma^9$

$\lambda(E5) = 2.4\times10^{-12}A^{10/3}E_\gamma^{11}$

$\lambda(M5) = 7.4\times10^{-13}A^{8/3}E_\gamma^{11}$

Physical Constants and Units

Acceleration due to gravity	g	9.81 m s^{-2}
Gravitational constant	G	6.674 $\times$ 10^{-11} N m^2 kg^{-2}
Ice point	T_{ice}	273.15 K
Avogadro constant	N_A	6.022 $\times$ 10^{23} mol^{-1}
		[*N.B.* 1 mole $\equiv$ 1 *gram-molecule*]
Gas constant	R	8.314 J K^{-1} mol^{-1}
Boltzmann constant	k, k_B	1.381 $\times$ 10^{-23} J K^{-1} $\equiv$ 8.62 $\times$ 10^{-5} eV K^{-1}
Stefan constant	σ	5.670 $\times$ 10^{-8} W m^{-2} K^{-4}
Rydberg constant	R_∞	1.097 $\times$ 10^{7} m^{-1}
	$R_\infty hc$	13.606 eV
Planck constant	h	6.626 $\times$ 10^{-34} J s $\equiv$ 4.136 $\times$ 10^{-15} eV s
$h/2\pi$	$\hbar$	1.055 $\times$ 10^{-34} J s $\equiv$ 6.582 $\times$ 10^{-16} eV s
Speed of light *in vacuo*	c	2.998 $\times$ 10^{8} m s^{-1}
	$\hbar c$	197.3 MeV fm
Charge of proton	e	1.602 $\times$ 10^{-19} C
Mass of electron	m_e	9.109 $\times$ 10^{-31} kg
Rest energy of electron		0.511 MeV
Mass of proton	m_p	1.673 $\times$ 10^{-27} kg
Rest energy of proton		938.3 MeV
One atomic mass unit	u	1.66 $\times$ 10^{-27} kg
Atomic mass unit energy equivalent		931.5 MeV
Electric constant	ϵ_0	8.854 $\times$ 10^{-12} F m^{-1}
Magnetic constant	μ_0	4π $\times$ 10^{-7} H m^{-1}
Bohr magneton	μ_B	9.274 $\times$ 10^{-24} A m^2 (J T^{-1})
Nuclear magneton	μ_N	5.051 $\times$ 10^{-27} A m^2 (J T^{-1})
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297 $\times$ 10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	2.426 $\times$ 10^{-12} m
Bohr radius	a_0	5.2918 $\times$ 10^{-11} m
angstrom	Å	10^{-10} m
barn	b	10^{-28} m^2
torr (mm Hg at 0 °C)	torr	133.32 Pa (N m^{-2})

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.